DPDP VAULT / DPDP ASSESSMENT PLATFORM

SOFTWARE COPYRIGHT REGISTRATION DOCUMENTATION

LITERARY WORK — COMPUTER SOFTWARE (INDIAN COPYRIGHT ACT, 1957)

DOCUMENT METADATA & COPYRIGHT PARTICULARS

Title of the Work:	DPDP Vault / DPDP Assessment Platform (Proposed Title)
Class of Work:	Literary Work — Computer Software (Section 2(o))
Author(s):	[APPLICANT AUTHOR NAME PLACEHOLDER — e.g. Full Legal Name]
Copyright Claimant:	[COPYRIGHT OWNER / ENTITY NAME PLACEHOLDER]
Programming Languages:	JavaScript (ES2024 / Node.js 20+), JSX (React 19), CSS3, JSON, MQL
Runtime Environment:	Next.js 15 App Router, Node.js 20+, Vercel Serverless Functions
Date of Completion:	[CREATION DATE PLACEHOLDER — e.g. October 2026]
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Country of Origin:	India

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1. INTRODUCTION AND PURPOSE OF THE SOFTWARE

DPDP Vault / DPDP Assessment Platform is an original computer software system engineered to automate the verification, auditing, and continuous compliance monitoring of digital websites and web applications under India's Digital Personal Data Protection Act, 2023 (DPDP Act) and Data Protection Board enforcement requirements.

The software addresses key industry challenges by replacing manual compliance audits with an autonomous, reproducible digital inspection framework:

- Empirical Digital Inspection: Automatically launches a sandboxed Chromium browser context to navigate to approved target domains, exploring interactive consent banners, privacy policies, grievance officer disclosures, and authentication forms.
- Statutory Reasoning via Multimodal AI: Integrates Google Gemini 3.8 Flash multimodal reasoning to visually comprehend and verify privacy notice accessibility, Schedule 8 Indian language provisions, and affirmative consent mechanics.
- Deterministic Scoring: Computes a standardized compliance percentage score (0–100%) weighted by statutory severity and applicability dates.
- Automated Compliance Reporting: Generates verifiable PDF compliance certificates and tabular Excel workbooks for technical remediation.

2. SYSTEM ARCHITECTURE AND EXECUTION MODEL

DPDP Vault implements a modern Direct Synchronous Serverless Architecture built upon the Next.js 15 App Router. The software eliminates the operational complexity of background daemons, worker instances, and distributed queue systems by encapsulating the entire inspection pipeline within high-memory serverless execution functions on Vercel.

Architecture highlights include:

- Next.js 15 App Router: Server Components and dynamic Route Handlers (/api/runs, /api/assessments/[id]/

reassess).

- Universal Browser Launcher: Environment-aware launcher dynamically utilizing @sparticuz/chromium on Vercel and Playwright Chromium locally.
- In-Memory Evidence Streaming: Screenshots are captured directly into memory buffers and uploaded to Cloudinary CDN, eliminating serverless disk bottlenecks.
- Mongoose / MongoDB Atlas: Document database storing 16 normalized schemas with cold-start pre-registration.

3. CORE FUNCTIONAL MODULES

3.1 Synchronous Assessment Engine (lib/assessment-runner.js)

Coordinates the end-to-end multi-stage inspection lifecycle across 7 statutory stages: Landing Page Notice (Sec 5), Affirmative Consent Banners (Sec 6), Grievance Redressal Officer Verification (Sec 8(9)), Consent Manager Granularity (Sec 6), Child Data Safeguards (Sec 9), Authentication Security (Sec 8(5)), and Privacy Document Analysis.

3.2 Canonical Statutory Requirement Registry (lib/requirement-registry.js)

An authoritative legal knowledge base encoding over 25 statutory rules from the DPDP Act, 2023 and Data Protection Board guidelines into machine-evaluable test specifications with stable identifiers, applicability dates, and scoring weights.

3.3 Compliance Scoring Engine (lib/assessment-engine.js & assessment-summary.js)

Computes normalized compliance scores using the formula: Score = (Passed Findings / Actionable Findings) * 100. Correctly filters out not_applicable and not_tested rules from the denominator to maintain mathematical integrity.

3.4 Multimodal AI Analysis (lib/gemini.js & lib/autonomous-agent.js)

Executes visual reasoning on captured screenshots using Google Gemini 3.8 Flash, evaluating language clarity, font readability, dark pattern absence, and statutory disclosures.

3.5 Security Controls & SSRF Defense (lib/security.js)

Implements multi-layer defensive filtering blocking loopbacks (127.0.0.0/8), private subnets (10/8, 172.16/12, 192.168/16), Carrier-Grade NAT (100.64.0.0/10), and cloud metadata endpoints (169.254.169.254). Encrypts credentials at rest using AES-256-GCM.

3.6 Evidence Redactor & Cloud Streaming (lib/cloudinary.js & lib/evidence-redactor.js)

Directly uploads screenshot buffers to Cloudinary CDN. Regex pattern matchers redact authorization headers, tokens, passwords, and sensitive keys from URLs, logs, and screenshots before persistence.

3.7 Report Generation Engine (lib/report-generator.js)

Compiles publication-grade PDF audit reports via PDFKit and detailed multi-tab Excel remediation workbooks via ExcelJS.

4. DATA MODELS & DATABASE SCHEMA SPECIFICATION

The platform defines 16 original Mongoose schemas in the models/ directory: User, Organization, ApprovedTarget, AssessmentProfile, Assessment, TestRun, Finding, LegalRequirement, ScenarioResult, Evidence, StructuredObservation, BrowserAction, AssessmentEvent, AssessmentDocument, AuditLog, and Report.

5. THIRD-PARTY OPEN SOURCE LIBRARIES

All compliance algorithms, inspection rules, and UI designs represent original proprietary work. The software utilizes the following permissive open-source packages as runtime infrastructure:

- Next.js 15.1.7 (MIT License) — Server-side framework
- React 19.0.0 (MIT License) — UI component library
- Mongoose 8.10.1 (MIT License) — MongoDB object modeling
- @google/genai 2.30.0 (Apache 2.0) — Google Gen AI SDK
- Playwright 1.64.1 & @sparticuz/chromium 153.0.0 (Apache 2.0 / MIT) — Browser automation
- PDFKit 0.20.2 & ExcelJS 4.4.0 (MIT License) — Report compilers
- bcryptjs 3.0.3 & jsonwebtoken 9.0.3 (MIT License) — Authentication security
- Tailwind CSS 3.4.17 (MIT License) — Stylesheet compiler

6. DECLARATION OF ORIGINAL AUTHORSHIP

The applicant affirms that DPDP Vault is an original computer software program (literary work) created through independent technical development. The proprietary software expression, architecture, and scoring models are the original work of the author(s).